Probability and Statistics

Chapter 5. Some Discrete Probability Distributions

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Chapter Overview

5.1 Introduction and Discrete Uniform distribution

5.1 Introduction and Discrete Uniform distribution

- A discrete probability distribution may be represented by a histogram, a table, or a formula.
- Many statistical experiments produce random variables with the same general behavior.

- **Discrete Uniform distribution**
- **Binomial distribution**
- **Hypergeometric distribution**
- **Negative binomial distribution**
- **Geometric distribution**
- **Poisson distribution**

Typical real-life interpretations

- Number of cured patients among treated patients $\rightarrow$ **Binomial**
- Number of defectives in a sample from a finite lot $\rightarrow$ **Hypergeometric**
- Number of trials until a false alarm or until a success occurs $\rightarrow$ **Geometric / Negative binomial**
- Number of white blood cells in a fixed amount of blood $\rightarrow$ **Poisson**

핵심은 공식을 외우는 것이 아니라, 상황의 구조를 보고 어떤 분포를 써야 하는지를 구분하는 것이다.

Discrete Uniform Distribution

A random variable X is called a **discrete uniform random variable** if it takes

$$x_1, x_2, \ldots, x_k$$

with equal probability.

If

$$P(X = x_i) = \frac{1}{k}, \qquad i = 1, 2, \ldots, k,$$

then X has a **discrete uniform distribution**.

- 모든 가능한 값이 같은 확률을 가진다.
- 유한한 개수의 outcomes가 equally likely 할 때 자연스럽게 등장한다.
- 예: fair die, equally likely tag numbers, random card labels

Mean and Variance Discrete Uniform Distribution

The mean and variance are

$$E(X) = \frac{1}{k}\sum_{i=1}^{k} x_i,$$

$$\mathrm{Var}(X) = \frac{1}{k}\sum_{i=1}^{k} x_i^2 - \left(\frac{1}{k}\sum_{i=1}^{k} x_i\right)^2.$$

Special Case: Consecutive Integers

If

$$X \in \{1, 2, \ldots, n\}$$

and all values are equally likely, then

$$P(X = x) = \frac{1}{n}, \qquad x = 1, 2, \ldots, n.$$

$$E(X) = \frac{n+1}{2}, \qquad \mathrm{Var}(X) = \frac{n^2 - 1}{12}.$$

- 평균은 가장 작은 값과 가장 큰 값의 중간이다.
- 분산 공식은 자주 쓰이므로 알아두면 편리하다.

Example 1: Fair Die

Let X be the number shown when a fair die is rolled once.

- Then X takes values $1, 2, 3, 4, 5, 6$.
- Since the die is fair,

$$P(X = x) = \frac{1}{6}, \qquad x = 1, 2, 3, 4, 5, 6.$$

Hence, X has a **discrete uniform distribution** on

$$\{1, 2, 3, 4, 5, 6\}.$$

Also,

$$E(X) = \frac{1+2+3+4+5+6}{6} = 3.5.$$

$$\mathrm{Var}(X) = \frac{6^2 - 1}{12} = \frac{35}{12}.$$

Example 2: Custom Finite Set

Suppose a random variable X takes the values $2, 5, 8, 11$ with equal probability.

Then

$$P(X = 2) = P(X = 5) = P(X = 8) = P(X = 11) = \frac{1}{4}.$$

So X is a discrete uniform random variable with $k = 4$.

Its mean is

$$E(X) = \frac{2 + 5 + 8 + 11}{4} = \frac{26}{4} = 6.5.$$

Its variance is

$$\mathrm{Var}(X) = \frac{2^2 + 5^2 + 8^2 + 11^2}{4} - (6.5)^2$$

$$= \frac{4 + 25 + 64 + 121}{4} - 42.25 = 53.5 - 42.25 = 11.25.$$

5.2 Binomial and Multinomial Distributions

Bernoulli Process

A **Bernoulli process** has the following properties:

1. The experiment consists of repeated trials.
2. Each trial results in either a **success** or a **failure**.
3. The probability of success, denoted by p, remains constant from trial to trial.
4. The repeated trials are independent.

If X denotes the number of successes in n Bernoulli trials, then X is called a **binomial random variable**.

Binomial Distribution

If a Bernoulli trial results in success with probability p and failure with probability $q = 1 - p$, then the probability distribution of the binomial random variable X, the number of successes in n independent trials, is

$$b(x; n, p) = \binom{n}{x} p^x q^{n-x}, \qquad x = 0, 1, 2, \ldots, n.$$

- $\binom{n}{x}$: number of arrangements of x successes in n trials
- $p^x q^{n-x}$: probability of one such arrangement

Where does the binomial formula come from?

Suppose we want exactly x successes in n independent trials.

- For one fixed order, the probability is

$$p^x q^{n-x}.$$

- The number of such orders is

$$\binom{n}{x}.$$

Therefore,

$$P(X = x) = \binom{n}{x} p^x q^{n-x}.$$

즉, 경우의 수와 한 경우의 확률를 곱한 것이다.

Why the name Binomial?

The binomial distribution is related to the expansion of

$$(q+p)^n.$$

Indeed,

$$(q+p)^n = \binom{n}{0}q^n + \binom{n}{1}pq^{n-1} + \binom{n}{2}p^2q^{n-2} + \cdots + \binom{n}{n}p^n.$$

Each term in the binomial expansion corresponds exactly to

$$b(x;n,p) = \binom{n}{x}p^xq^{n-x}.$$

Hence,

$$\sum_{x=0}^{n} b(x;n,p) = 1.$$

Example 5.1

The probability that a certain kind of component will survive a shock test is $3/4$. Find the probability that exactly 2 of the next 4 components tested survive.

Let X be the number of components that survive. Then

$$X \sim \mathrm{Bin}(4, 3/4).$$

Hence,

$$P(X = 2) = b\left(2; 4, \frac{3}{4}\right) = \binom{4}{2}\left(\frac{3}{4}\right)^2\left(\frac{1}{4}\right)^2 = \frac{27}{128}.$$

Example 5.2

The probability that a patient recovers from a rare blood disease is 0.4. If 15 people are known to have contracted this disease, what is the probability that

(a) at least 10 survive?

(b) from 3 to 8 survive?

(c) exactly 5 survive?

Example 5.2: Solution (1)

Let X be the number of people who survive. Then

$$X \sim \mathrm{Bin}(15, 0.4).$$

(a)

$$P(X \geq 10) = 1 - P(X < 10) = 1 - \sum_{x=0}^{9} b(x; 15, 0.4).$$

From the table,

$$P(X \geq 10) = 1 - 0.9662 = 0.0338.$$

(b)

$$P(3 \leq X \leq 8) = \sum_{x=3}^{8} b(x; 15, 0.4).$$

$$P(3 \leq X \leq 8) = \sum_{x=0}^{8} b(x; 15, 0.4) - \sum_{x=0}^{2} b(x; 15, 0.4).$$

$$= 0.9050 - 0.0271 = 0.8779.$$

Example 5.2: Solution (2)

(c)

$$P(X = 5) = b(5; 15, 0.4).$$

Using cumulative binomial probabilities,

$$P(X = 5) = \sum_{x=0}^{5} b(x; 15, 0.4) - \sum_{x=0}^{4} b(x; 15, 0.4).$$

$$= 0.4032 - 0.2173 = 0.1859.$$

Example 5.3

A large chain retailer purchases a certain kind of electronic device from a manufacturer. The manufacturer indicates that the defective rate of the device is 3%.

(a) The inspector randomly picks 20 items from a shipment. What is the probability that there will be at least one defective item among these 20?

(b) Suppose that the retailer receives 10 shipments in a month and the inspector randomly tests 20 devices per shipment. What is the probability that there will be exactly 3 shipments each containing at least one defective device among the 20 that are selected and tested from the shipment? (소매 체인점이 한 달에 10번의 출하를 받고, 검사관이 출하된 제품마다 무작위로 20개의 기기를 검사한다. 이 경우, 선하된 제품에서 선택되어 검사된 20개 제품 중 적어도 하나가 불량인 출하가 정확히 3번일 확률은 얼마인가?)

Example 5.3: Solution

(a) Let X be the number of defectives among the 20. Then

$$X \sim \mathrm{Bin}(20, 0.03).$$

Hence,

$$P(X \geq 1) = 1 - P(X = 0) = 1 - b(0; 20, 0.03)$$
$$= 1 - (0.97)^{20} = 0.4562.$$

(b) For each shipment, define success = "at least one defective among the 20 tested." Then

$$p = 0.4562.$$

If Y is the number of such shipments among 10, then

$$Y \sim \mathrm{Bin}(10, 0.4562). \quad => \quad P(Y = 3) = \binom{10}{3}(0.4562)^3(1 - 0.4562)^7 = 0.1602.$$

Theorem 5.1

The mean and variance of the binomial distribution $b(x; n, p)$ are

$$\mu = np, \qquad \sigma^2 = npq,$$

where $q = 1 - p$.

Theorem 5.1: Proof

Let the outcome on the jth trial be represented by an indicator variable

$$I_j = \begin{cases} 1, & \text{if trial } j \text{ is a success,} \\ 0, & \text{if trial } j \text{ is a failure.} \end{cases}$$

Then

$$X = I_1 + I_2 + \cdots + I_n.$$

Since

$$E(I_j) = 0 \cdot q + 1 \cdot p = p,$$

we have

$$E(X) = E(I_1) + \cdots + E(I_n) = np.$$

Also,

$$\mathrm{Var}(I_j) = E(I_j^2) - [E(I_j)]^2 = p - p^2 = p(1-p) = pq.$$

Since the trials are independent,

$$\mathrm{Var}(X) = \mathrm{Var}(I_1) + \cdots + \mathrm{Var}(I_n) = npq.$$

Example 5.4

It is conjectured that an impurity exists in 30% of all drinking wells in a certain rural community. Ten wells are randomly selected for testing.

(a) Using the binomial distribution, what is the probability that exactly 3 wells have the impurity, assuming that the conjecture is correct?

(b) What is the probability that more than 3 wells are impure?

Let $X \sim \mathrm{Bin}(10, 0.3)$.

$$\text{(a)} \quad P(X = 3) = \binom{10}{3}(0.3)^3(0.7)^7 \approx 0.2668.$$

$$\text{(b)} \quad P(X > 3) = 1 - P(X \leq 3) \approx 0.3504.$$

Example 5.5

Find the mean and variance of the binomial random variable of Example 5.2, and then use Chebyshev's theorem to interpret the interval $\mu \pm 2\sigma$.

Example 5.5: Solution

In Example 5.2,

$$X \sim \mathrm{Bin}(15, 0.4).$$

$$\mu = np = (15)(0.4) = 6, \qquad \sigma^2 = npq = (15)(0.4)(0.6) = 3.6.$$

$$\sigma = \sqrt{3.6} \approx 1.897.$$

Hence,

$$\mu \pm 2\sigma = 6 \pm 2(1.897).$$

So the interval is

$$(2.206,\ 9.794).$$

By Chebyshev's theorem, the number of recoveries among 15 patients has probability at least $3/4$ of falling between 2.206 and 9.794, or because the data are discrete, roughly between 2 and 10 inclusive.

Example 5.6

Consider the situation of Example 5.4. Suppose 10 wells are randomly selected and 6 are found to contain the impurity.

What does this imply about the conjecture that 30% of all wells are impure? Use a probability statement.

Example 5.6: Solution

If the conjecture is correct, then

$$X \sim \mathrm{Bin}(10, 0.3).$$

We ask:

$$P(X \geq 6) = 1 - P(X \leq 5).$$

Using the cumulative binomial probabilities,

$$P(X \geq 6) = 1 - 0.9527 = 0.0473.$$

Thus, if only 30% of all wells are impure, then the chance of observing 6 or more impure wells out of 10 is only 0.0473. This is quite unlikely, so the data cast considerable doubt on the conjecture and suggest that the impurity problem may be more severe. (따라서 전체 우물 중 30% 만 불순물이 포함되어 있다면, 10개 우물 중 6개 이상에서 불순물이 발견될 확률은 이론적으로 0.0473에 불과하다. 이는 매우 가능성이 낮은데 실제로 6개에서 발견되었으므로 가정이 잘못되었을 수 있음을 시사한다.)

Multinomial Distribution

The binomial experiment becomes a **multinomial experiment** if each trial has more than two possible outcomes.

If a trial can result in any one of the k outcomes

$$E_1, E_2, \ldots, E_k$$

with probabilities

$$p_1, p_2, \ldots, p_k, \qquad \sum_{i=1}^{k} p_i = 1,$$

then the joint probability that

$$X_1 = x_1,\ X_2 = x_2,\ \ldots,\ X_k = x_k$$

in n independent trials is given by the multinomial distribution.

Multinomial Distribution Formula

If a given trial can result in the k outcomes $E_1, \ldots, E_k$ with probabilities $p_1, \ldots, p_k$, then

$$f(x_1, x_2, \ldots, x_k; p_1, p_2, \ldots, p_k, n) = \frac{n!}{x_1! x_2! \cdots x_k!} p_1^{x_1} p_2^{x_2} \cdots p_k^{x_k},$$

where

$$x_1 + x_2 + \cdots + x_k = n$$

and

$$p_1 + p_2 + \cdots + p_k = 1.$$

Why the multinomial formula works

- One specific arrangement with counts $x_1, \dots, x_k$ has probability

$$p_1^{x_1} p_2^{x_2} \cdots p_k^{x_k}.$$

- The number of such arrangements is

$$\frac{n!}{x_1! x_2! \cdots x_k!}.$$

따라서 multinomial distribution도 binomial과 같은 논리로 유도된다:

$$\text{number of arrangements} \times \text{probability of one arrangement.}$$

Example 5.7

For a certain airport with three runways, the probabilities that the individual runways are accessed by a randomly arriving commercial jet are

$$p_1 = \frac{2}{9}, \qquad p_2 = \frac{1}{6}, \qquad p_3 = \frac{11}{18}.$$

What is the probability that 6 randomly arriving airplanes are distributed as follows?

- Runway 1: 2 airplanes
- Runway 2: 1 airplane
- Runway 3: 3 airplanes

Example 5.7: Solution

Let X_1, X_2, X_3 denote the numbers of airplanes assigned to runways 1, 2, and 3. Then

$$P(X_1 = 2, X_2 = 1, X_3 = 3) = \frac{6!}{2!1!3!} \left(\frac{2}{9}\right)^2 \left(\frac{1}{6}\right) \left(\frac{11}{18}\right)^3.$$

Hence,

$$P(X_1 = 2, X_2 = 1, X_3 = 3) = 0.1127.$$

5.3 Hypergeometric Distribution

Hypergeometric Distribution

The key difference between the binomial and hypergeometric distributions is the sampling scheme.

- **Binomial**: sampling with replacement or independent trials
- **Hypergeometric**: sampling without replacement

A hypergeometric experiment has two properties:

1. A random sample of size n is selected without replacement from N items.
2. Of the N items, k are classified as successes and $N - k$ as failures.

Hypergeometric Distribution Formula

If X is the number of successes in the sample, then:

$$h(x; N, n, k) = \frac{\binom{k}{x}\binom{N-k}{n-x}}{\binom{N}{n}}, \qquad \max\{0, n-(N-k)\} \leq x \leq \min\{n, k\}.$$

- Total number of samples of size n from N: $\binom{N}{n}$
- Favorable samples with exactly x successes: $\binom{k}{x}\binom{N-k}{n-x}$

Example 5.8

A particular part used as an injection device is sold in lots of 10. The producer deems a lot acceptable if no more than one defective is in the lot.

A sampling plan involves randomly selecting and testing 3 parts out of 10. If none of the 3 is defective, the lot is accepted.

Comment on the utility of this plan.

(주사 장치로 사용되는 특정 부품이 10개 단위로 판매된다. 생산자는 해당 묶음에 불량품이 1개 이하인 경우 그 묶음을 합격으로 간주한다. 샘플링 계획은 10개 중 3개를 무작위로 선택하여 테스트하는 것이다. 3개의 부품 모두 불량품이 아니면 해당 로트는 합격 처리된다. 이 계획의 유용성에 대해 논하시오.)

Example 5.8: Solution

Assume the lot is truly unacceptable, that is, 2 out of 10 parts are defective.

Let X be the number of defectives found in the sample of 3. Then

$$X \sim \mathrm{Hypergeo}(10, 3, 2).$$

The lot is accepted if $X = 0$:

$$P(X = 0) = \frac{\binom{2}{0}\binom{8}{3}}{\binom{10}{3}} = 0.467.$$

Thus, even when the lot is truly unacceptable, it will be accepted about 46.7% of the time. So this sampling plan should be considered faulty.

Example 5.9

Lots of 40 components each are deemed unacceptable if they contain 3 or more defectives. The procedure for sampling a lot is to select 5 components at random and reject the lot if a defective is found.

What is the probability that exactly 1 defective is found in the sample if there are 3 defectives in the entire lot?

Example 5.9: Solution

Here,

$$N = 40, \qquad n = 5, \qquad k = 3, \qquad x = 1.$$

Thus,

$$P(X = 1) = h(1; 40, 5, 3) = \frac{\binom{3}{1}\binom{37}{4}}{\binom{40}{5}} = 0.3011.$$

So this plan detects a bad lot containing 3 defectives only about 30% of the time.

Theorem 5.2

The mean and variance of the hypergeometric distribution $h(x; N, n, k)$ are

$$\mu = n\frac{k}{N},$$

$$\sigma^2 = \frac{N-n}{N-1} \cdot n \cdot \frac{k}{N}\left(1 - \frac{k}{N}\right).$$

The factor

$$\frac{N-n}{N-1}$$

is called the **finite population correction**. 비복원추출에서는 independence가 없으므로 binomial보다 분산이 작아진다.

Example 5.10

A lot of 100 items contains 12 defectives. What is the probability that in a sample of 10, exactly 3 are defective?

Example 5.10: Solution

Let X be the number of defectives in the sample. Then

$$X \sim \mathrm{Hypergeo}(100, 10, 12).$$

Hence,

$$P(X = 3) = h(3; 100, 10, 12) = \frac{\binom{12}{3}\binom{88}{7}}{\binom{100}{10}} = 0.08.$$

Example 5.11

Find the mean and variance of the random variable of Example 5.9 and then use Chebyshev's theorem to interpret the interval $\mu \pm 2\sigma$.

Example 5.11: Solution

For Example 5.9,

$$N = 40, \qquad n = 5, \qquad k = 3.$$

$$\mu = n\frac{k}{N} = 5 \cdot \frac{3}{40} = \frac{3}{8} = 0.375.$$

$$\sigma^2 = \left(\frac{40-5}{39}\right)(5)\left(\frac{3}{40}\right)\left(1 - \frac{3}{40}\right) = 0.3113.$$

$$\sigma = \sqrt{0.3113} \approx 0.558.$$

Thus,

$$\mu \pm 2\sigma = 0.375 \pm 2(0.558) = 0.375 \pm 1.116,$$

so the interval is

$$(-0.741,\ 1.491).$$

Chebyshev's theorem implies that the number of defectives obtained when 5 components are selected at random from a lot of 40 containing 3 defectives has probability at least 3/4 of falling in this interval. Since the number of defectives must be an integer, this means that at least three-fourths of the time the sample contains fewer than 2 defectives.

Relationship to the Binomial Distribution

If the sample size n is small compared with the population size N, then sampling without replacement behaves approximately like independent sampling.

Rule of thumb:

$$\frac{n}{N} \le 0.05$$

이면 hypergeometric distribution을 binomial distribution으로 근사해도 좋다.

In that case,

$$p = \frac{k}{N}$$

plays the role of the binomial success probability.

Example 5.12

A manufacturer of automobile tires reports that among a shipment of 5000 sent to a local distributor, 1000 are slightly blemished. If one purchases 10 of these tires at random, what is the probability that exactly 3 are blemished?

Example 5.12: Solution

Here,

$$N = 5000, \qquad k = 1000, \qquad n = 10, \qquad p = \frac{1000}{5000} = 0.2.$$

Since

$$\frac{n}{N} = \frac{10}{5000} = 0.002,$$

the binomial approximation is excellent.

$$h(3; 5000, 10, 1000) \approx b(3; 10, 0.2) = \binom{10}{3}(0.2)^3(0.8)^7 = 0.2013.$$

The exact probability is 0.2015, so the approximation is very accurate.

Multivariate Hypergeometric Distribution

Suppose the N items can be partitioned into k cells

$$A_1, A_2, \ldots, A_k$$

with

$$a_1, a_2, \ldots, a_k$$

elements, respectively.

If a random sample of size n is selected, and X_i is the number of sampled elements from A_i, then the joint distribution is called the **multivariate hypergeometric distribution**.

Multivariate Hypergeometric Formula

$$f(x_1, x_2, \ldots, x_k; a_1, a_2, \ldots, a_k, N, n) = \frac{\binom{a_1}{x_1}\binom{a_2}{x_2}\cdots\binom{a_k}{x_k}}{\binom{N}{n}},$$

where

$$x_1 + x_2 + \cdots + x_k = n$$

and

$$a_1 + a_2 + \cdots + a_k = N.$$

Example 5.13

A group of 10 individuals is used for a biological case study. The group contains

- 3 people with blood type O,
- 4 people with blood type A,
- 3 people with blood type B.

What is the probability that a random sample of 5 will contain

$$\text{1 person with type O}, \quad \text{2 people with type A}, \quad \text{2 people with type B?}$$

Example 5.13: Solution

$$P = \frac{\binom{3}{1}\binom{4}{2}\binom{3}{2}}{\binom{10}{5}} = \frac{3 \cdot 6 \cdot 3}{252} = \frac{3}{14}.$$

5.4 Negative Binomial and Geometric Distributions

Negative Binomial Distribution

In a negative binomial experiment, the trials continue until a fixed number of successes occurs.

If repeated independent trials can result in success with probability p and failure with probability $q = 1 - p$, then the probability distribution of the random variable X, the trial on which the kth success occurs, is

$$b^*(x; k, p) = \binom{x-1}{k-1} p^k q^{x-k}, \qquad x = k, k+1, k+2, \ldots$$

Why the negative binomial formula works

To have the kth success on trial x:

- trial x must be a success;
- among the first $x - 1$ trials, there must be exactly $k - 1$ successes and $x - k$ failures.

Thus,

$$P(X = x) = \binom{x-1}{k-1} p^k q^{x-k}.$$

Example 5.14

In an NBA championship series, the team that wins 4 games out of 7 is the winner. Suppose teams A and B face each other and team A has probability 0.55 of winning a game over team B.

(a) What is the probability that team A will win the series in 6 games?

(b) What is the probability that team A will win the series?

(c) If the two teams were facing each other in a regional playoff series decided by winning 3 out of 5 games, what is the probability that team A would win the series?

Example 5.14: Solution (1)

(a) For team A to win the series in 6 games, the 4th success must occur on game 6:

$$P(X = 6) = b^*(6; 4, 0.55) = \binom{5}{3}(0.55)^4(0.45)^2 = 0.1853.$$

(b) Team A can win the championship in 4, 5, 6, or 7 games:

$$P(\text{A wins}) = b^*(4; 4, 0.55) + b^*(5; 4, 0.55) + b^*(6; 4, 0.55) + b^*(7; 4, 0.55).$$

$$= 0.0915 + 0.1647 + 0.1853 + 0.1668 = 0.6083.$$

Example 5.14: Solution (2)

(c) In a best-of-5 series, team A needs 3 wins:

$$P(\text{A wins}) = b^*(3; 3, 0.55) + b^*(4; 3, 0.55) + b^*(5; 3, 0.55).$$

$$= 0.1664 + 0.2246 + 0.2021 = 0.5931.$$

Geometric Distribution

The geometric distribution is the special case of the negative binomial distribution with $k = 1$.

If repeated independent trials can result in success with probability p and failure with probability $q = 1 - p$, then the probability distribution of the random variable X, the number of the trial on which the first success occurs, is

$$g(x; p) = pq^{x-1}, \qquad x = 1, 2, 3, \ldots$$

Example 5.15

For a certain manufacturing process, it is known that, on the average, 1 in every 100 items is defective. What is the probability that the fifth item inspected is the first defective item found?

Example 5.15: Solution

Here,

$$p = 0.01, \qquad q = 0.99.$$

So

$$P(X = 5) = g(5; 0.01) = (0.01)(0.99)^4 = 0.0096.$$

Example 5.16

At a busy time, a telephone exchange is very near capacity, so callers have difficulty placing their calls. Suppose that the probability of a connection during a busy time is $p = 0.05$.

What is the probability that 5 attempts are necessary for a successful call?

Example 5.16: Solution

Let X be the number of attempts required until the first successful connection. Then

$$X \sim \mathrm{Geom}(0.05).$$

Hence,

$$P(X = 5) = g(5; 0.05) = (0.05)(0.95)^4 = 0.041.$$

Theorem 5.3

The mean and variance of a geometric random variable are

$$\mu = \frac{1}{p}, \qquad \sigma^2 = \frac{1-p}{p^2}.$$

Geometric distribution은 “first success까지의 waiting time”을 나타내므로, p가 작을수록 평균 waiting time $1/p$가 커진다.

5.5 Poisson Distribution and the Poisson Process

Poisson Process

Experiments that count the number of outcomes occurring during a given time interval or in a specified region are called **Poisson experiments**.

Examples:

- number of telephone calls received per hour
- number of days school is closed due to snow
- number of bacteria in a culture
- number of typing errors per page

Properties of the Poisson Process

A Poisson process has the following properties:

1. The number of outcomes in one interval or region is independent of the number in any other disjoint interval or region.
2. The probability of one outcome in a very short interval is proportional to the length of the interval.
3. The probability of more than one outcome in a very short interval is negligible.

쉽게 말하면, 희귀 사건이 시간/공간 위에 흩어져 발생하는 모델이다.

Poisson Distribution

If X is the number of outcomes in a given interval or region of size t, and the average rate is λ per unit, then X follows a Poisson distribution with mean λt.

$$p(x; \lambda t) = e^{-\lambda t} \frac{(\lambda t)^x}{x!}, \qquad x = 0, 1, 2, \ldots$$

Example 5.17

During a laboratory experiment, the average number of radioactive particles passing through a counter in 1 millisecond is 4. What is the probability that 6 particles enter the counter in a given millisecond?

Example 5.17: Solution

Let $X \sim \mathrm{Poisson}(4)$. Then

$$P(X = 6) = p(6; 4) = e^{-4}\frac{4^6}{6!}.$$

Using the table,

$$P(X = 6) = \sum_{x=0}^{6} p(x; 4) - \sum_{x=0}^{5} p(x; 4) = 0.8893 - 0.7851 = 0.1042.$$

Example 5.18

Ten is the average number of oil tankers arriving each day at a certain port. The facilities at the port can handle at most 15 tankers per day.

What is the probability that on a given day tankers have to be turned away?

Example 5.18: Solution

Let X be the number of tankers arriving in one day. Then

$$X \sim \mathrm{Poisson}(10).$$

Tankers have to be turned away if

$$X > 15.$$

So

$$P(X > 15) = 1 - P(X \leq 15).$$

Using the Poisson table,

$$P(X > 15) = 1 - 0.9513 = 0.0487.$$

Theorem 5.4

Both the mean and the variance of the Poisson distribution $p(x; \lambda t)$ are

$$\mu = \lambda t, \qquad \sigma^2 = \lambda t.$$

Shape of the Poisson Distribution

As the mean μ grows larger, the Poisson distribution becomes more symmetric and bell-shaped.

For small μ, the distribution is strongly right-skewed.
For larger μ, the distribution begins to resemble a symmetric distribution.

Poisson Approximation to the Binomial

The Poisson distribution can be viewed as a limiting form of the binomial distribution.

If

$$X \sim \mathrm{Bin}(n, p),$$

with

$$n \to \infty, \qquad p \to 0, \qquad np \to \mu,$$

then

$$b(x; n, p) \to p(x; \mu).$$

즉, 시행 수는 많고 success probability는 매우 작을 때,

$$\mathrm{Bin}(n, p) \approx \mathrm{Poisson}(np).$$

Why the approximation is useful

When n is large and p is small, direct binomial calculations may be cumbersome.

- Binomial parameters: (n, p)
- Poisson approximation parameter: $\mu = np$

Rare-event counting problems are often easier to compute with the Poisson approximation.

Example 5.19

In a certain industrial facility, accidents occur infrequently. It is known that the probability of an accident on any given day is 0.005, and accidents are independent of each other.

(a) What is the probability that in a given period of 400 days there will be an accident on one day?

(b) What is the probability that there are at most three days with an accident?

Example 5.19: Solution

Let

$$X \sim \mathrm{Bin}(400, 0.005).$$

Then

$$np = (400)(0.005) = 2.$$

Use the Poisson approximation

$$X \approx \mathrm{Poisson}(2).$$

(a)

$$P(X = 1) \approx e^{-2}\frac{2^1}{1!} = 0.271.$$

(more precisely 0.2707)

(b)

$$P(X \leq 3) \approx \sum_{x=0}^{3} e^{-2}\frac{2^x}{x!} = 0.857.$$

Example 5.20

In a manufacturing process where glass products are made, defects or bubbles occur occasionally. It is known that, on average, 1 in every 1000 items produced has one or more bubbles.

What is the probability that a random sample of 8000 will yield fewer than 7 items possessing bubbles?

Example 5.20: Solution

This is essentially a binomial experiment with

$$n = 8000, \qquad p = 0.001.$$

Since p is very small and n is large, use the Poisson approximation with

$$\mu = np = (8000)(0.001) = 8.$$

If X is the number of bubbly items, then

$$P(X < 7) = P(X \leq 6) \approx \sum_{x=0}^{6} e^{-8} \frac{8^x}{x!} = 0.3134.$$

5.6 Potential Misconceptions and Hazards

5.6 Potential Misconceptions and Hazards

Although these distributions are widely useful, they can be misused if the assumptions are not checked carefully.

- In practical applications, the parameter values are often estimated, not known exactly.
- In binomial models, the assumptions of independence and constant p may fail.
- In Poisson models, the event rate may change over time.
- Historical data may not reflect the current state of the process.

핵심은, **distribution formula**를 적용하기 전에 **assumptions**가 맞는지 먼저 확인하라는 것이다.

Important practical warning

A famous misuse of the binomial distribution occurred in the 1961 baseball season, when probability calculations were used to predict which player would break Babe Ruth's home-run record.

The major mistake was estimating the parameter p from historical career frequency without accounting for the changing conditions of the current season. 즉, model 자체보다도 **parameter choice and model assumptions**가 더 큰 문제일 수 있다.